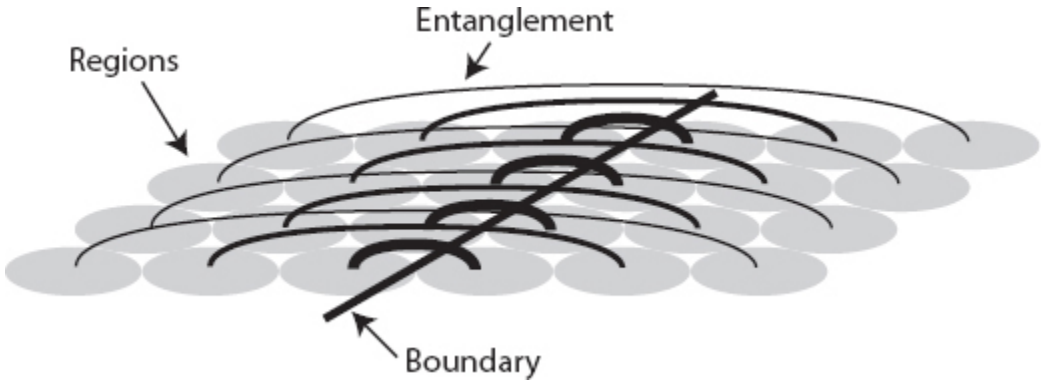
sounds vaguely reminiscent of Einstein's equation, which also connects geometry (the curvature of spacetime) to a matter quantity (energy). Are they somehow related?



They could be, as was pointed out in a provocative 1995 paper by Ted Jacobson, an ingenious physicist at the University of Maryland. In ordinary quantum field theory without gravity, entropy is proportional to area in the vacuum state, but in higher-energy states it doesn't have to be. Jacobson postulated that there's something special about gravity: when gravity is included, the entropy of a region is *always* proportional to its boundary area. That's not at all what we would expect in quantum field theory, but maybe it happens once gravity enters the game. We can imagine that it might be the case, and see what happens.

What happens is pretty wonderful. Jacobson posited that the area of a surface is proportional to the entropy of the region it encloses. Area is a geometric quantity; we can't calculate the area of a surface without knowing something about the geometry of the space it is a part of. Jacobson noted that we could relate the area of a very small surface to the same geometric quantity that appears on the left-hand side of Einstein's equation. Meanwhile, entropy tells us something about "matter," broadly construed; about the stuff that is living within spacetime. The concept of entropy originally arose within thermodynamics, where it was related to the heat leaving a system. And heat is a form of energy. Jacobson also argued that this entropy could be directly related to the energy term appearing on the right-hand side of Einstein's equation. Through these maneuvers he was able to *derive* Einstein's equation for general relativity, rather than directly postulating it, as Einstein did.